FORM-C

COUNCIL OF SCIENTIFIC AND INDUSTRIAL RESEARCH
Human Resource Development Group
CSIR Complex, Library Avenue, Pusa, New Delhi – 110012

(This application for financial support by CSIR to the research proposal should be typed; Twenty copies are to be submitted /forwarded to the Head, HRDG at the above address)

PART A

1.Institute to administer the grant	Indian Institute of Information Technology, Allahabad, Uttar Pradesh, India

*Send recognition/affiliation certificate in case of private institutes.

2. Project Title (use not more than four lines) :	Modeling and Simulation of Environmental and Occupational Epidemiology data in Indian context: An Ontology based Data Mining Approach

3. General area of the proposed research (Please ✓ the subject area).		
☐ Animal Sciences & Biotechnology ☐ Inorganic and Physical Chemistry ☐ Organic and Medicinal Chemistry ☐ Disaster Preparedness	☐ Earth Sciences ☐ Mathematical Sciences ☐ Physical Sciences	☐ Engineering Sciences √ Medical Sciences ☐ Plant Sciences

4. Name of the sponsoring CSIR Laboratory (if applicable) with address and name of concerned scientist with telephone, fax, e-mail	Not Applicable

5. Principal Investigator (PI):

a. Title : Dr. b. Name : Sonali Agarwal	Sex: Female
c. Full Official Address	Dr. Sonali Agarwal Assistant Professor, IT Department Room No.: 5203, CC3 Building Indian Institute of Information Technology, Allahabad Devghat, Jhalwa, Allahabad -211015, (UP) INDIA
Mobile/Telephone Fax E-mail	Phone: 0532-2922424, Mobile:09415647042 91-532-2922125, sonali@iiita.ac.in
d. Position	Assistant Professor
e. Date of birth	May 04, 1976
f. Highest Degree University/Institute Date	Ph.D. in Information Technology, Indian Institute of Information Technology, Allahabad, August, 2011
g. Total time to be devoted to project (in man months per year)	3 Years

6. Other participants (give name, address, and highest qualification for each of the Co-Principal-Investigator) (CO-I):

Er AH Khan, M Tech Principal Scientist, Environmental Monitoring Laboratory, CSIR-Indian Institute of Toxicology Research, Gheru campus, Sarojini Nagar Industrial campus, Kanpur road, Amausi post office, Lucknow 226008, Uttar Pradesh, India. Email: ahkhan@iitr.res.in Alternate Email: ahkhan2005@gmail.com Tel: 05222476228	Dr. C. Kesavachandran, PhD Principal Scientist, CSIR-Indian Institute of Toxicology Research, Epidemiology Laboratory Vishvigyan Bhavan, 31, Mahatma Gandhi Marg, Lucknow 226001, Uttar Pradesh, India. Email: ckchandran@iitr.res.in Alternate Email: ckesavachandran@gmail.com Tel: 0522-2627586 Extension: 683

7. Names and addresses of four research scientists actively engaged in the general area of the proposed research:

(a) Expert in Environmental and Occupational Epidemiology Dr AK Srivastava, MD Former Director, National Institute of Miners Health, Nagpur; Rtd. Senior Principal Scientist, CSIR-IITR, Lucknow 104, Rohtas Enclave, Ravindrapalli, Faizabad Road, Lucknow - 226016	(b) Expert in Epidemiological Statistics Mr Neeraj Mathur Retd. Senior Principal Scientist, CSIR-IITR, Lucknow D1/110, Sector F, Jankipuram, Lucknow 226021, UP
(c) Expert in Data Mining Prof. K. K. Shukla Department of Computer Science, Indian Institute of Technology, (BHU), Varanasi +91-542-230-7056 Kkshukla.cse@iiitbhu.ac.in	(d) Expert in Software Engineering Prof. A. K. Agrawal, (Retd.) Department of Computer Science Indian Institute of Technology (BHU), Varanasi B-36/40, Kabir Nagar Varanasi 211005 +91-532-9451940493 akagrawal.iitbhu@gmail.com

8. Research support availed/being availed/applied for by the PI from different sources, including CSIR, during the last six years: NIL

Grant agency	Title of the project and reference number	Duration(from mm/yy to mm/yy)	Percentage of time devoted /being devoted/to be devoted, in man months	Amount in lakh Rs.
To be filled				

9. Proposed budget:

Budget items	Amount requested in Rs.		
	1st Year	**2nd Year**	**3rd Year**
(a) Manpower			
Staff JRF (IITA)	**3,00,000**	**3,00,000**	**3,36,000**
(b) Contingency			
Travel (IITA) Travel (CSIR-IITR)	**50,000** **50,000**	**50,000** **50,000**	**1,00,000** **50,000**
Budget for Consumables (IITA) Budget for consumables (CSIR-IITR)	**50,000** **50,000**	**50,000** **50,000**	**50,000** **50,000**
Other Costs/Contingency Cost Other Costs/Contingency Cost (CSIR-IITR)	**50,000** **50,000**	**50,000** **50,000**	**50,000** **50,000**
Institute Over Heads	**1,50,000**	**1,50,000**	**1,00,000**
(c) Equipment (item wise)			
Servers	**4, 30,000**	**NIL**	**NIL**
Client Machines	**3,00,000**	**NIL**	**NIL**
Switches & Networking Equipments	**50,000**	**NIL**	**NIL**
Printer Printer with scanner (CSIR-IITR)	**14,000** **20,000**	**NIL**	**NIL**
Scanner	**6,000**	**NIL**	**NIL**
(d) Total	**15,70,000**	**7,50,000**	**7,86,000**
GRAND TOTAL	**31,06,000**		

10. Available institutional facilities:

IIIT-Allahabad	CSIR-IITR
1. Workshop Facility: Full time available 2. Water & Electricity: Full Time available 3. Laboratory Space/ Furniture: Full Time available 4. Power Generator: Full Time available 5. AC Room or AC: Full Time available 6. Telecommunication including e-mail & fax: available 7. Transportation: Yes (Sharing Basis) 8. Administrative/ Secretarial support: available 9. Information facilities like Internet/Library: Full Time available 10. Computational facilities: Full Time available	1. Repository for data base management of occupational and environmental health data at CSIR-IITR 2. Epidemiological Statistical analysis expertise available 3. Experts available for Occupational and Environmental epidemiology 4. All the other infrastructure facility to conduct the study

11. Declaration and attestation:

We certify that all the details declared here are correct and complete. 1. Signature of PI	Date:
2. Signature of CO-Is	Date:
3. Signature of CO-Is	Date:

12. Certificate of the heads of the department and institution:

<table>
<tr><td colspan="2">We have read the terms and conditions of the CSIR Extramural Research Grant Scheme. The necessary institutional facilities are available and will be provided for the implementation of this research proposal being submitted to the CSIR for funding. Full account of expenditure will be rendered by the institution yearly.</td></tr>
<tr><td>Name of the head :
of the department

Signature with date :

Seal :</td><td>Name of the head :
of the institution

Signature with date :

Seal :</td></tr>
</table>

In case of CSIR Sponsored Scheme:

<table>
<tr><td colspan="2">We have read the terms and conditions of the CSIR Extramural Research Grant Scheme. The necessary institutional facilities are available and will be provided for the implementation of this research proposal being submitted to the CSIR for funding. Full account of expenditure will be rendered by the institution yearly.</td></tr>
<tr><td>Name of the head :
of the department

Signature with date :

Seal :</td><td>Name of the head :
of the institution

Signature with date :

Seal :</td></tr>
<tr><td colspan="2">Forwarded by the
Director of the
CSIR Laboratory

Signature with Date :

Seal:</td></tr>
</table>

PART B

DETAILED RESEARCH PROPOSAL

Your description should be comprehensive and brief and should in no case exceed 10 pages **(pages in excess will be discarded).**

Information to be given below must illustrate the research ground to be covered through this project. The research objectives and the scientific perspective against which the project has been conceived should be clearly mentioned to enable the Referees to arrive at a scientific judgment.

Give the particulars in the format shown below and send 20 copies of the entire project to the following address:

Head, HRDG
CSIR Complex
Library Avenue, Pusa
New Delhi 110012

13. Title of the project:
"Modeling and Simulation of Environmental and Occupational Epidemiology data in Indian context: An Ontology based Data Mining Approach"

14. Aims and significance of the project
(Include the current status of work in area, both in India and abroad, with appropriate reference list at the end; identify lacunae, define question to be investigated; list briefly specific objectives of investigation. Ethical clearance be enclosed where necessary).

Current Status in India
In a recently reported research, it is identified that uncontrolled urbanization and industrialization is becoming a serious concern of health care-givers because these changes are leading to environmental degradation and appearing as a root cause of cancers, neurodegenerative diseases, cardio-respiratory diseases, gastro-intestinal disorders, genito-urinary disorders. In the latest Global Burden of Diseases (GBD) 2015 updates, household pollution remained among the leading three risks for DALYs across 800 geographies [1]. Similarly, ambient particulate matter pollution ranked the second leading risk in India. The worst scenario of unsafe water was declared the sixth –leading risk for DALYs in India [2]. In the competitive era of global market and continuously evolving business needs have affected the occupational morbidity drastically. Healthcare industries are constantly upgrading but still the gap exists to eliminate the chronic diseases such as heart disease, cancer and stress related problems. At the same time due to lack of awareness about occupational safety and environmental hazards processionals are bound to work in unhealthy environment [3]. As per World Health Organization data occupational health risks are recognized as a tenth leading cause of morbidity and mortality [4]. The disease burden due to occupational risk has share about 1.5% of the global burden in terms of Disability Adjusted Life Years (DALY). Out of overall risk commonly reported heath problems and associated risks include approximately 37% of backache, 16% of hearing loss, 13% of lung disease, 11% of asthma, 10% of injuries, 9% of cancer and 2% of leukemia [4].

It is also evident that severities of occupational health risk are more prominent in some communities especially in low income groups. Due to unavailability of central and standard repositories of epidemiological data, the occupational health situation in India is not well recognized and understood by the policy makers. Director General of Factory Advisory Services and Labor Institutes (DGFASLI) raised the need of such linkages so that initiatives could be taken at global level to overcome the shortcomings of present situations [5]. In an another study, Leigh et al., have found incidence of work-related illness between 924,700 and 1,902,300; and 121,000 casualties in India every year. [6]. The Indian Council of Medical Research reported that the workers of coal mines and textile industries are suffering from serious diseases such as pneumoconiosis and byssinosis [7] [8]. Similarly the occupational health risk due to asbestosis, silicosis, lead poisoning, carbon disulfide and manganese poisoning are also being reported by many agencies [9]. The presence of such diseases female working population can led to certain concerns such as adverse effects on reproduction [10]. It is also evident from the research that occupational and environmental health care issues could not be separated and must be handled together for fighting against disease burden [11]. Although several action plans are in place to minimize the effects of chemicals, in view of their health effects, not much have been changing globally on this issue. There are very few professional agencies like National Institute of Occupational Health (NIOH), Regional Occupational Health Center (ROHC), National Occupational Health Center (NOHC), CSIR-Indian Institute of Toxicology Research and Central Labor Institute are working in the area of work-related hazards since independence, in India.

The lack of proper nationwide database related to health and environment may lead to misconceptions and contradictions among the policy planners. The strong interrelationship of environmental health linkages with socio-cultural issues, financial credibility, literacy among public, sanitation and hygiene, industrial and domestic waste emissions, dietary and personal habits, public health issues etc make the understanding of this cause effect relationship of environmental and occupational health more complex. Ontology development is a recent area of research which consist domain specific entities along with properties and relationships [12]. Although many domain specific ontology present in India, there are very limited researches available about ontology development in the area of Environmental and Occupational Epidemiology [13][14]. Ontology development using environmental and occupational epidemiology data especially in Indian scenario will simplify and structure the modeling and simulation of occupational and environmental healthcare issues[15][16][17].

Current Status in Abroad

Occupational illness is a slow progressing mechanism and could be noticed only over a period of time. Majority of time, it is due to poor hygiene, unsafe exposure to physical, chemical or biological agents and disease-causing bacteria and viruses. The extreme cases of occupational illness may lead to mechanical, chemical, biological, physical, and psychosocial hazards. Around 2.9 billion workers all over the world suffer from this problem. According to World Health Organization (WHO) and the International Labour Organization (ILO), there are around 140,000 to 355,000 deaths have been recorded per year [6, 18-20] due to occupational illness [19, 21, 22]. According to WHO the status of occupational services is also not satisfactory and only 20% to 50% of workers of countries having industrial background, and less than 10% of workers belonging to developing countries, have privilege avail occupational health services [23].

As per International Labour organization following occupational health services are essential to be provided: (1) Occupation risk assessment and identification, (2) action plan for workplace hazards and survival, (3) safe and clean workplace design, (4) over-all upgradation of work practices, (5) standards for occupational health, safety, and hygiene, (6) Monitoring of workers' health, (7) providing comfortable adaptation of work to the worker, (8) Occupational rehabilitation programs, (9) conducting training and awareness programs, (10) availability of emergency services and first aid, and (11) preventive measures to avoid injury and illness [9]. The above services could be only possible if the government agencies stand with strong will, policies and competence. Providers of occupational health services may be attracted if essential infrastructural support could be extended to them. It is also essential to adopt Information and Communication (ICT) practices in order to support occupational hygienists, safety specialists, occupational physicians, nurses, toxicologists, ergonomists, and epidemiologists so that better services could be performed by them. The ILO, WHO, and academic and nongovernmental organizations (NGOs) have already convinced that occupational health care can be improved with the help of recently developed practices such as Data Mining, Big Data, Social Media Mining and Ontology based solutions [24,25]. Ontology for Occupational Health was developed in 2011 in order to address the occupational risk. It consist structured representation of workplace healthcare, safety, security and environmental issues [26]. In an another case study, LinKBase® and LinKFactory® ontologies have been further extended to occupational healthcare [27][28]. The epidemiology ontology and Infectious Disease Ontology exist but they are being utilized for data mning operation. Existing ontologies can be used for data mining operations if we are designing them in structured way with precise relationship among entities. [29][30].

Lacunae in the present system

- Although environmental and occupational exposure related health problems are very common in India, few studies are conducted in a systematic manner incorporating epidemiological standard operating procedures.
- The above epidemiological studies have resulted in a better understanding of the occupational and environmental exposure related problems, but fail to build up a database repository for occupational and environmental health in India.
- There is a genuine need to develop epidemiological data analytics and models of environmental and occupational disease especially in Indian scenario to frame guidelines and policies for better environment, work place and health.

Questions to be investigated

- What is the association between personals habits viz., dietary/ smoking pattern and disease burden in Indian population?
- What is the association between body composition factors and Cardio-respiratory problems in Indian population?
- How different types of exposure (pesticides, ambient and indoor air) and its duration can cause different type of diseases in Indian scenario?
- Is there any association between pesticide exposure and health problems?
- Is there any relationship between outdoor air pollutants and cardio-respiratory problems in Indian population?
- What is the impact of Indoor air pollutants on cardio-respiratory and kidney problems?

Specific objective of Investigation

Establishment of an Ontology driven knowledge base to implement standardized occupational and environmental health based models by using Data Mining techniques for the assessment of health conditions during exposure of toxicants at workplaces/environment and develop appropriate model structures for toxicity related health problems in India.

15. Plan of work, methods and techniques to be used

The existing data generated from three studies viz., pesticide exposure and health problems in pesticide sprayers, ambient air pollution and its health effects in general population and indoor air pollutants and its impact on health among kitchen workers will be sorted, cleaned and used for the further analysis. All the above studies are approved by Institutional Human Ethics committee (details enclosed). If there is any missing data or lack of data available in a specific group, gender wise deficiencies in existing data etc, an epidemiological survey will be undertaken to include the missing data for the betterment of data analytics and as per the requirement of ontology based data mining approach. If any further epidemiological survey will be conducted, then approval will be taken from Institutional Human Ethics committee before the commencement of the study. In this research program we propose to develop Ontology driven knowledge base to support data mining processes, specifically designed for occupational and environmental health. The proposed framework consists of mathematical models based on epidemiological data, preventive models for occupational/ environmental health problems, rule based systems for occupational & environment health and forecasting & alert generation systems for health problems based on pesticide toxicity, indoor and outdoor air pollutants. Epidemiological data considered for the domain specific ontology development will be related to dietary pattern, smoking pattern, body composition factors, residential as well as other specific workplace exposure, outdoor and indoor air pollutants etc. The framework development will be done on following phases:

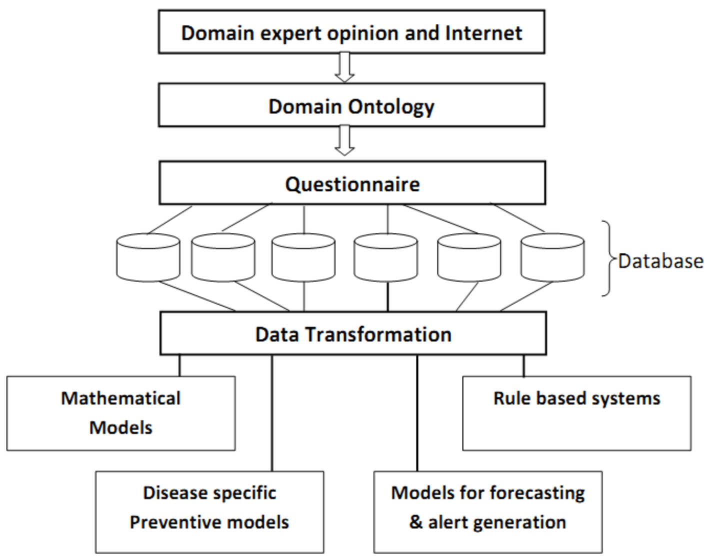

Figure 1: Proposed framework

Phase 1: Domain understanding

Since occupational and environmental healthcare is a concern of everybody now it is important to identify specific domains for complete monitoring and control. This research work will be mainly aimed on developing a fundamental consciousness about occupational and environmental health in general. For broader coverage of the stated research objective, an in-depth analysis of following case studies will be undertaken.

Case study 1: Association between dietary/ smoking pattern and disease burden

Case study 2: Association between body composition factors and Cardio-respiratory problems

Case study 3: Exposure duration at residential/work place and its related variations in disease patterns

Case study 4: Outdoor air pollutants and cardio-respiratory problems

Case study 5: Indoor air pollutants with cardio-respiratory and kidney related problems

Phase 2: Development of Epidemiological Ontology

Epidemiological Ontology is a way of structuring information related to epidemiology. It is a collection of standardized well defined epidemiological terms which are interconnected with logical relationship and designed to serve as knowledge generation engine with special emphasis in this project proposal on occupational and environmental health in Indian scenario. It supports standardized vocabulary with accurate data integration and enables complex queries in order to resolve issues of taxonomy, granularity and specificity. Following are the steps of Ontology development which will serve as a backbone for modeling of epidemiological data.

- **Step 1: Knowledge acquisition and conceptualization.**

To achieve a high coverage of epidemiological concepts, detailed requirements have to be collected. It could help to understand technical specifications of the system under development. It is important to collect structured and tractable data about the domain which should include cross-references, definitions, list of synonyms, persistent identifiers etc.

- **Step 2: Literature curation pipeline**

In this phase all potential documents related to environmental and occupational healthcare will be indentified. In this phase we collect concepts and terms about Epidemiology data and their interconnection to occupational and environmental health will be established with the help of linguistic pre-processing and named-entity recognition.

- **Step 3: Development of Ontology**

In this step all event occurrences must be identified. Commonly occurred objects in any environmental and occupational epidemiological events are “exposure”, “duration”, “morbidity” “symptoms”, “lung functions”, “microalbuminuria” etc. Possible relationship such as “being_exposed”, “being_morbid” and other concepts may also be listed down as per domain information available. Object such as “spread” is also required to be added.

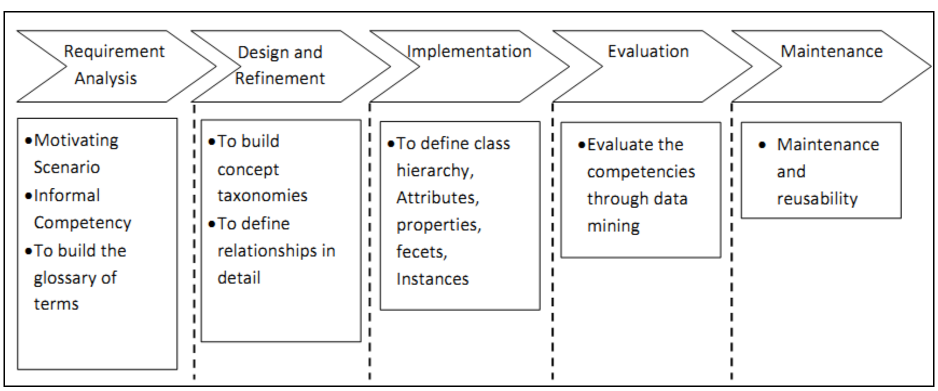

Figure 2: Ontology Development Process

A formal definition of all objects, relationships and behavior must be identified with the help of domain experts for modeling of occupational and environmental diseases. The various steps involved in ontology development are shown in Figure 2.

- **Step 4: Verification and validation of Ontology**

The verification of an ontology will be followed by taxonomy evaluation and generate information about inconsistency, incompleteness and redundancy about the available concept definitions on the other hand the validation process will help to ensure consistency, completeness, conciseness, expandability and sensitiveness.

Phase 3: Development of Mathematical models using Epidemiological data
Mathematical models are helpful for quantitative representation of environmental and occupational epidemiological data processes and its consequences as various disease outbreaks. Therefore it is essential to identify detailed mathematical representations of the indispensable features of the epidemiological variables at multiple scales, including location and temporal details, in an attempt to establish realistic models of environmental and occupational epidemiological parameters which will comprise a large set of ordinary differential equations (ODEs) describing time-varying epidemiological variables and parameters and its impact on various issues related to occupation and environmental health.

Phase 4: Ontology based Data Mining for Occupational and Environmental Health
This phase is mainly divided into three modules as indicated in figure 3. In module one preventive modeling for environmental and occupational diseases has been covered considering its large scale health risk impact on the general population exposed to air pollution and workers exposed to indoor air pollutants and pesticides etc Descriptive Data Mining techniques such as clustering and associated mining can be performed to identify the problem and suggest preventive measures for better well being. Module two will be designed for generation of rule based systems based on Epidemiological parameters based on the concept of decision trees. Forecasting and Alert Generating System for effective Occupational and Environmental Health will be included in module three and could be achieved by using classification and prediction techniques of data mining.

16. Time-table or milestones

<u>BAR diagram of Phase Wise Time Chart</u>

Activities	Duration in Months																	
	1-2	**3-4**	**5-6**	**7-8**	**9-10**	**11-12**	**13-14**	**15-16**	**17-18**	**19 -20**	**21-22**	**23-24**	**25-26**	**27-28**	**29-30**	**31-32**	**33-34**	**35-36**
Phase 1																		
Phase 2																		
Phase 3																		
Phase 4																		

17. Deliverables (Apart from reports/papers; identify any products, systems, hardware, software, synthesised compound etc. to be delivered at the end of the project)

Following may be the expected outcomes/deliverables:

i) A website for the project under development
ii) Ontology driven knowledge base for Occupational/Environmental epidemiology data and a Data Warehouse for Query Based operations, policy makers, researchers, global studies etc.

iii) Mathematical models based on occupational and environmental epidemiological data for understanding the problem and effective intervention of preventive tools.
iv) Disease specific preventive models for occupational and environmental toxicants.
v) Rule based systems based on occupational and environmental epidemiological parameters.
vi) Health alert system based on the toxic levels of pollution for effective Occupational and Environmental Health problems for the industrial workers and general public.

TABLE 1: List of deliverables

Deliverable ID	Phase ID	Deliverable Name	Details/features/Ingredients	Month Deadline
D.1.1	Phase one (P1)	Domain understanding and project website development	A web site for project	M4
			A open source web repository for all resources without any payment mode	
D.2.1	Phase two P2	Development of an Ontology driven knowledge base for Occupational/ Environmental health	Occupational/Environmental health Knowledge acquisition and conceptualization	M6
			Literature Curation Pipe line	M8
			Construction of ontology	M14
			Verification and validation of ontology	M16
D.3.1	Phase Three P3	Mathematical modeling of Occupational/Environmental health data.	Assumptions, parameter selection and model development	M24
			Model Comparison	
			Risk Management	
D.4.1	Phase Four P4	Preventive Models for Occupational/ Environmental health problems	Preventive modeling for environmental and occupational diseases	M28
D.4.2			Generation of rule based systems based on occupational and environmental epidemiological parameters	M32
D.4.3			Forecasting health alert generating system for effective occupational and environmental health problems	M36

18. Justification of Budget

(For each position, item of equipment, and contingencies. Quotation for equipment should also be enclosed)

Justification for Manpower

One Junior Research Fellow (JRF) with M.Tech or equivalent is required to do the detailed literature survey, development of algorithms and to perform the experiments for testing of the proposed techniques. This post will be the pay of Rs. 25,000/- per month (as per DST office memorandum No.SB/S9/Z-01/2015 dt. 07.1.2015) for first two years.

In third year, JRF will be promoted to Senior Research Fellow (SRF) based on his performance and pay of Rs. 28,000/- per month will be given as salary.

Justification for Equipments

i. It is required for research support.
ii. Required for Program execution and data computation at client side.
iii. It is required for high performance, better scalability, higher memory and storage requirements of Distributed Processing environment especially in case of large multimedia data.
iv. It is a dense rack server that is a reliable fit for virtualization, databases or high performance requirements of Distributed Processing environment.
v. It is required for research support.
vi. Required for the preparation of distributed environment and experimental setup
vii. Printers with scanner at CSIR-IITR for print out of excel sheets, documents and scan of documents related to study.

Justification for Consumables

During the time of experimental setup, there will be need for some storage media like CDs and DVDs for data storage purpose. Also there will be the regular requirements of papers, printer cartridges, etc. CSIR-IITR. Requirements of stationeries, printer cartridges, storage media etc.

Justification for Travel

It will be required to visit some places to meet the people working in the similar area and discuss the issues of the proposal. Cost requirement will increase in third year because of attending conferences/seminars/ workshops/symposium for sharing the result with others. Similar to above also required for CSIR-IITR investigators relevant to occupational and environmental epidemiology context.

Justification for Contingency

Contingencies are kept for books, online resources etc. Similar to above also required for CSIR-IITR investigators relevant to occupational and environmental epidemiology context.

Justification for Overhead

Overheads may exist and taken according to yearly expenses. May be only your institute may quote here?

19. If the project has any industrial significance, give names and addresses of 3 industries that may be interested in the project.

20. If sponsored by CSIR Laboratory please give detail of technical programme that will be carried out at the CSIR laboratory and name, date of birth & designation of the CSIR scientist collaborating and recommendation letter from the Director of collaborating laboratory.

21. If applicable, indicate any link with ongoing projects of the sponsoring CSIR laboratory.

22. List not more than 25 of your publications with full bibliographic details/reports/patents or other documents in the last 10 years (Use asterisks to identify publications relevant to this proposal).

Selected Publications of Dr. Sonali Agarwal, Principal Investigator

1. Divya Tomar, **Sonali Agarwal**, “Twin Support Vector Machine for Multiple Instance Learning based on Bag Dissimilarities”, minor revision required in Advances in Artificial Intelligence **(Hindawi Journal).**
2. Divya Tomar, **Sonali Agarwal,** "Leaf Recognition for Plant Classification using Direct Acyclic Graph Based Multi-Class Least Squares Twin Support Vector Machine" accepted in International Journal of Image and Graphics, 2016.
3. Divya Tomar, **Sonali Agarwal,** “Prediction of Defective Software Modules Using Class Imbalance Learning” Published in Applied Computational Intelligence and Soft Computing, Hindawai, Volume 2016, Article ID 7658207, 12 pages, http://dx.doi.org/10.1155/2016/7658207
4. Divya Tomar, **Sonali Agarwal** “Multiclass Least Squares Twin Support Vector Machine for Pattern Classification” published in International Journal of Data Base and Application (IJDTA) Volume 8, Issue 6 Pages pp.285-302 http://dx.doi.org/10.14257/ijdta.2015.8.6.26.
5. Divya Tomar, **Sonali Agarwal** "An effective Weighted Multi-class Least Squares Twin Support Vector Machine” for Imbalanced data classification" published in International Journal of Computational Intelligence Systems, Taylor & Francis, Vol. 8, issue 4, July 2015, Pages 761–771. **SCI Journal, Impact Factor=0.574**
6. Divya Tomar, **Sonali Agarwal** "A comparison on multi-class classification methods based on least squares twin support vector machine" published in Knowledge-Based Systems, Elsevier, Vol. 81, June 2015, Pages 131–147. SCI Journal, Scopus indexed, Impact Factor=3.011
7. Divya Tomar, **Sonali Agarwal** "Twin Support Vector Machine: A review from 2007 to 2014"published in Egyptian Informatics Journal, Elsevier, Vol. 16, issue 1, March 2015, Pages 55–69. **Scopus indexed, Impact Factor= 0.810**
8. Divya Tomar, Divya Ojha, **Sonali Agarwal** "An Emotion Detection System Based on Multi Least Squares Twin Support Vector Machine" published in Advances in Artificial Intelligence, Hindawi Publishing Corporation, Vol. 2014, Article ID 282659, 11 pages, 2014. doi:10.1155/2014/282659.
9. Divya Tomar, **Sonali Agarwal** "Hybrid Feature Selection Based Weighted Least Squares Twin Support Vector Machine Approach for Diagnosing Breast Cancer, Hepatitis, and Diabetes" published in Advances in Artificial Intelligence, Hindawi Publishing Corporation, Vol. 2015, Article ID 265637, 10 pages, 2015. doi:10.1155/2015/265637
10. **Sonali Agarwal,** Divya Tomar "A Feature Selection Based Model for Software Defect Prediction" published in International Journal of Advanced Science and Technology Vol. 65 (2014), pp.39-58 http://dx.doi.org/10.14257/ijast..2014.65.04
11. Divya Tomar, **Sonali Agarwal**, "A Survey of Data Mining approaches for Healthcare" published in International Journal of Bio-Science and Bio-Technology Vol.5, No.5 (2013), pp. 241-266 http://dx.doi.org/10.14257/ ijbsbt.2013.5.5.25. **Scopus Indexed**
12. DivyaTomar, Sonali Agarwal, "Multiple instance learning based on Twin Support Vector Machine" accepted in International Conference on Computer, Communication and Computational Science, IC4S, 2016, Springer, Ajmer, India.
13. DivyaTomar, Sonali Agarwal, “A Multilabel approach using Binary Relevance and One-versus-Rest Least Squares Twin Support Vector Machine for Scene Classification” is presented in the International Conference on Computational intelligence and communication technology (ICICT-2016), Ghaziabad, February 12-13.

14. DivyaTomar, Sonali Agarwal, “Multi-class Twin Support Vector Machine for Pattern Classification " presented in 3rd International Conference on Advanced Computing, Networking, and Informatics (ICACNI – 2015) at School of Computer Engineering, Kalinga Institute of Industrial Technology, (KIIT University), Orissa, India 23 - 25 June 2015. Acceptance Rate : 25%, Scopus indexed
15. DivyaTomar, Sonali Agarwal, “Multi-label Classifier for Emotion Recognition from Music "presented in 3rd International Conference on Advanced Computing, Networking, and Informatics (ICACNI – 2015) at School of Computer Engineering, Kalinga Institute of Industrial Technology, (KIIT University), Orissa, India 23 - 25 June 2015. Acceptance Rate : 25%, Scopus indexed
16. DivyaTomar, Sonali Agarwal, "Direct acyclic graph based multi-class twin support vector machine for pattern classification " In Proceedings of the Second ACM IKDD Conference on Data Sciences (CoDS '15). ACM, New York, NY, USA, 80-85. DOI=10.1145/2732587.2732598,http://doi.acm.org/10.1145/2732587.2732598

Selected Publications of Dr. A, H. Khan Co- Principal Investigator

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2. GBD 2015 Disease and Injury Incidence and Prevalence Collaborators, and others (CN Kesavachandran). Global, regional, and national incidence, prevalence, and years lived with disability for 310 diseases and injuries, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet*, 2016, Vol. 388, No. 10053 **[IF: 44]**.
3. GBD 2015 DALYs and HALE Collaborators, and others (CN Kesavachandran). Global, regional, and national disability-adjusted life-years (DALYs) for 315 diseases and injuries and healthy life expectancy (HALE), 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet*, 2016 Vol. 388, No. 10053 **[IF: 44]**.
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23. List of awards and honors conferred on the PI with dates.

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